

$$P\left[X = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{bmatrix} \mid H_j\right] = \prod_{i=1}^n e^{-\lambda_j} \frac{\lambda_j^{k_i}}{k_i!} = \left[e^{-\lambda_j} \right]^n \frac{\lambda_j^{\sum_{i=1}^n k_i}}{\prod_{i=1}^n (k_i!)}$$

$$P\left[X = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{bmatrix} \mid H_1\right] > P\left[X = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{bmatrix} \mid H_2\right]$$

$$\begin{aligned} & \left(e^{-\lambda_j} \frac{\lambda_j^{k_1}}{k_1!} \right) \left(e^{-\lambda_j} \frac{\lambda_j^{k_2}}{k_2!} \right) \left(e^{-\lambda_j} \frac{\lambda_j^{k_3}}{k_3!} \right) \dots \left(e^{-\lambda_j} \frac{\lambda_j^{k_n}}{k_n!} \right) \\ &= e^{-n\lambda_j} \frac{\lambda_j^{k_1+k_2+\dots+k_n}}{k_1! k_2! \dots k_n!} \end{aligned}$$

$$e^{-n\lambda_1} \frac{\lambda_1^{k_1+k_2+\dots+k_n}}{\cancel{k_1! k_2! \dots k_n!}} > e^{-n\lambda_2} \frac{\lambda_2^{k_1+k_2+\dots+k_n}}{\cancel{k_2! k_2! \dots k_n!}}$$

$$\frac{\lambda_1^{k_1+k_2+\dots+k_n}}{\lambda_2^{k_1+k_2+\dots+k_n}} > e^{n(\lambda_1-\lambda_2)}$$

$$\left(\frac{\lambda_1}{\lambda_2} \right)^{k_1+k_2+\dots+k_n} > e^{n(\lambda_1-\lambda_2)}$$

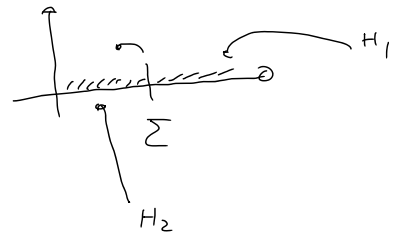
$$\ln \left[\left(\frac{\lambda_1}{\lambda_2} \right)^{k_1+k_2+\dots+k_n} \right] > \ln \left[e^{n(\lambda_1-\lambda_2)} \right]$$

$$(k_1+k_2+\dots+k_n) \ln \left(\frac{\lambda_1}{\lambda_2} \right) > n(\lambda_1-\lambda_2)$$

$$k_1+k_2+\dots+k_n \stackrel{H_1}{>} \frac{n(\lambda_1-\lambda_2)}{\ln \left(\frac{\lambda_1}{\lambda_2} \right)} = \sum = \frac{-n(\lambda_2-\lambda_1)}{-\ln \left(\frac{\lambda_2}{\lambda_1} \right)}$$

$$k_1 + k_2 + \dots + k_n > \frac{\dots}{\ln\left(\frac{x_1}{x_2}\right)} = \angle = \frac{\dots}{-\ln\left(\frac{x_2}{x_1}\right)}$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \rightarrow \boxed{} \quad Z: X_1 + X_2 + \dots + X_n > \Sigma$$



$$\underbrace{x_1 + x_2 + x_3}_{w_1 + x_3}$$

$$Z = \underbrace{x_1 + x_2 + x_3}_{w_1} + x_4$$

$$E[Z] = E[X_1] + E[X_2] + \dots + E[X_n]$$

$$E[Z | H_j] = \sum_{n > j} \dots \quad P[Z = k | H_j] = \frac{(n > j)^k}{k!}$$

Z è poss.

$$P(E) = \underbrace{P(H_2)}^{\frac{1}{2}} \underbrace{P(E|H_2)}^{\frac{1}{2}} + \underbrace{P(H_1)}^{\frac{1}{2}} \underbrace{P(E|H_1)}^{\frac{1}{2}}$$

$$= \frac{1}{2} \left[P[Z > \Sigma | H_2] + P[Z \leq \Sigma | H_1] \right]$$

$$P[Z \leq \Sigma | H_1] = F_{Z/H_1}(\Sigma)$$

$$P[Z > \Sigma | H_2] = 1 - P[Z \leq \Sigma | H_2] = 1 - F_{Z/H_2}(\Sigma)$$

$$F_x^{(n)} = \sum_{k=0}^n e^{-\lambda} \frac{\lambda^k}{k!} = \underbrace{\sum_{k=0}^{+\infty} e^{-\lambda} \frac{\lambda^k}{k!}}_1 - \sum_{k=n+1}^{+\infty} e^{-\lambda} \frac{\lambda^k}{k!}$$

$$X(n) \in \{0, 1\} \quad \text{i.i.d.}$$

$$P[X(n)=1] = p$$

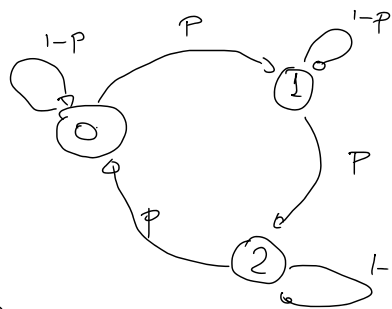
$$Y(n) \in \{0, 1, 2\}$$

$$Y(n) = [Y(n-1) + X(n)] \bmod 3$$

$$Y(n) \mid Y(n-1)=2 = \left[2 + X(n) \right] \bmod 3$$

$$g[Y(n-1), X(n)]$$

$$Y^{(n)} / Y^{(n-1)} = 2$$



$$(P^T - \lambda I) \underline{x} = \underline{0}$$

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 1-P & P & 0 \\ 0 & 1-P & P \\ P & 0 & 1-P \end{bmatrix} \end{matrix} \quad \underline{x} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1-P-\lambda & 0 & P & 0 & 0 & 0 \\ P & 1-P-\lambda & 0 & 0 & 0 & 0 \\ 0 & P & 1-P-\lambda & 0 & 0 & 0 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} -P & 0 & P \\ P & -P & 0 \\ 0 & P & -P \end{bmatrix}}_{(P^T - \lambda I)} \underbrace{\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix}}_{\underline{x}} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{\underline{0}}$$

$$(1-P-\lambda) (1-P-\lambda)^2 + P P^2 = 0$$

$$\begin{cases} -P P_2 + P P_3 = 0 \Leftrightarrow P_1 = P_3 \\ P P_1 - P P_2 = 0 \Leftrightarrow P_1 = P_2 \end{cases}$$

$$(1-P-\lambda)^3 = -P^3$$

$$1-P-\lambda = -P$$

$$\{P_1 = P_2 = P_3\}$$

$$\lambda = 1$$

$$P_1 + P_2 + P_3 = 1$$

$$3P_1 = 1 \quad P_1 = \frac{1}{3} = P_2 = P_3$$

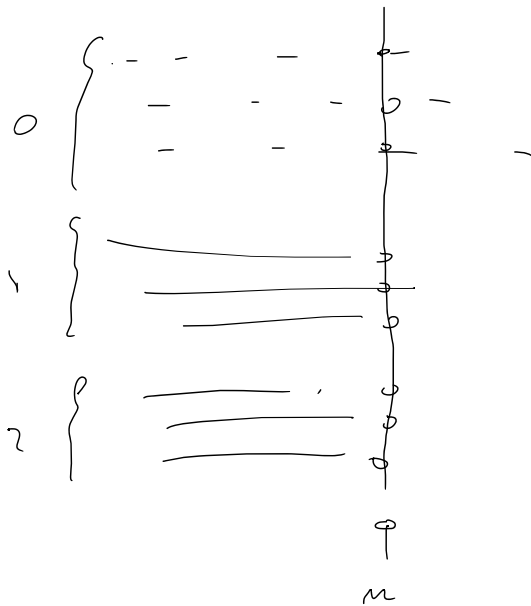
$$Y(0) = 0, 1, 2$$

for $i = 1 : 10000$

$$Y(i) = [Y(i-1) + \{0, 1, 2\}]_{\text{mod } 3}$$

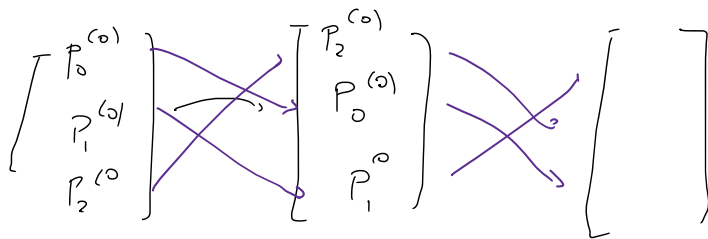
end

$$\underline{Y} = [Y(0), Y(1), \dots, Y(10000)]$$



$$\begin{bmatrix} P_0^{(1)} \\ P_1^{(1)} \\ P_2^{(1)} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} P_0^{(0)} \\ P_1^{(0)} \\ P_2^{(0)} \end{bmatrix}$$

$$\begin{aligned} P_0^{(1)} &= P_2^{(0)} \\ P_1^{(1)} &= P_0^{(0)} \\ P_2^{(1)} &= P_1^{(0)} \end{aligned}$$



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$$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \rightarrow \begin{matrix} H(10) \\ H(11) \\ H(12) \end{matrix}$$

$$P[Y(n-1)=0] H[Y(n) | Y(n-1)=0] + P[Y(n-1)=1] H[Y(n) | Y(n-1)=1] + P[Y(n-1)=2] H[Y(n) | Y(n-1)=2]$$

$$\sum_i P_i \lg \frac{1}{P_i}$$

$$P_i = 2^{-n} \quad \frac{1}{P_i} = 2^n \quad \lg\left(\frac{1}{P_i}\right) = n$$

$$P_i \lg \frac{1}{P_i} = n 2^{-n} \xrightarrow{n \rightarrow \infty} 0$$

$$\begin{cases} X_n \in \{0, 1, 2\} & P[X(n)=0] = 1 \\ & P[X(n)=1] = P[X(n)=2] \end{cases}$$

$$P(E|H_0) \equiv \text{PROB. DI FALSO ALLARME}$$

$$P(E) = P(H_0) P(E|H_0) + P(H_1) P(E|H_1)$$

$$P(E|H_1) = 1 - \underbrace{P[C|H_1]}_{\text{prob. di rilevazione} = \text{potenza del test}}$$

$$E|_{H_0} \equiv \text{FALSO ALLARME}$$

$$\begin{cases} \max P[C|H_1] \\ P[E|H_0] \leq \alpha \end{cases}$$

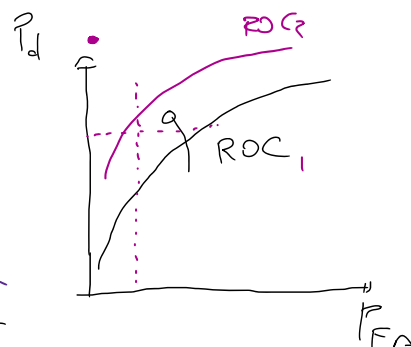
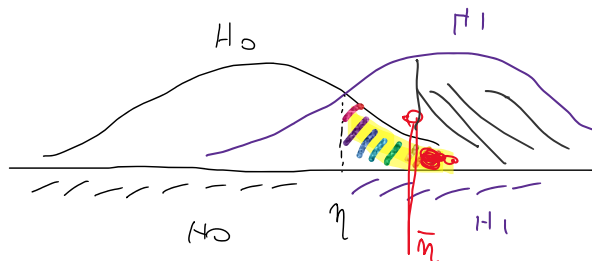
$$\frac{f_{\underline{X}}(x|H_1)}{f_{\underline{X}}(x|H_0)}$$

$$\{x_1, x_2, \dots, x_n\}$$

$$\mathcal{L}[\underline{X}] = \begin{cases} \frac{P[\underline{X} = x_i | H_1]}{P[\underline{X} = x_i | H_0]} > \eta \end{cases}$$

$$\boxed{\eta = 1}$$

$$P[E|H_0] = P[\mathcal{L}(\underline{X}) > \eta | H_0] = P[$$



$$P[E|H_0] = P[X > \eta | H_0] = P[X_{|H_0} > \eta] = \int_{\eta}^{+\infty} f_{X|H_0}(x) dx$$

$$P[C|H_1] = P[X > \eta | H_1] = P[X_{|H_1} > \eta] = \int_{\eta}^{+\infty} f_{X|H_1}(x) dx$$

$$\frac{\hat{\theta}}{\sqrt{g(\cdot)}} \rightarrow \hat{\theta}$$

$$\int 2(\hat{\theta} - \theta) f_{\theta/x_n}(\theta) d\theta = 0$$

$$\hat{\theta} \underbrace{\int f_{\theta/x_n}(\theta) d\theta}_{=1} = \int \theta f_{\theta/x_n}(\theta) d\theta = E[\theta | x_n]$$

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